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# Cipher Decryption - Assessment

Graded Assignment • 15 min

Due Nov 23, 11:59 PM IST

## Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

Next item →

**1.** Why is the log of probabilities used instead of the probabilities themselves? 1 / 1 point

- Logs are easier to communicate to end users
- It provides no real advantage, just style
- Multiplying probabilities can lead to underflows
- Logs are not actually used, just probabilities

✓ Nice work

Correct! Logs prevent underflow by converting multiplication into addition.

**2.** Why are genetic algorithms used in deciphering messages? 1 / 1 point

- They can perfectly map DNA sequences to letters
- They can efficiently explore large solution spaces
- They only focus on predetermined solutions
- They decode messages instantly

✓ Nice work

Correct! Genetic algorithms are efficient for exploring large spaces of possible solutions.

**3.** What is the role of the initial mapping in encoding? 1 / 1 point

- It checks cipher text accuracy
- It defines how plain text converts to cipher text
- It randomly assigns characters
- It directly decodes messages

✓ Nice work

Correct! Initial mapping is foundational for encoding properly.

**4.** What do DNA strings represent in the context of cryptographic genetic algorithms? 1 / 1 point

- They store biological data
- They are placeholders for final solutions
- They represent potential mappings
- They are hardcoded into the algorithm

✓ Nice work

Correct! DNA strings symbolize potential mappings or solutions.

**5.** What is the difference between unigram and bigram probabilities? 1 / 1 point

- Unigrams consider single probabilities; bigrams consider pairs
- Both are the same, just different terms
- Unigrams use two elements; bigrams use one
- Unigrams are for longer texts; bigrams are for shorter texts

✓ Nice work

Correct! Unigrams deal with single probabilities; bigrams with pairs.